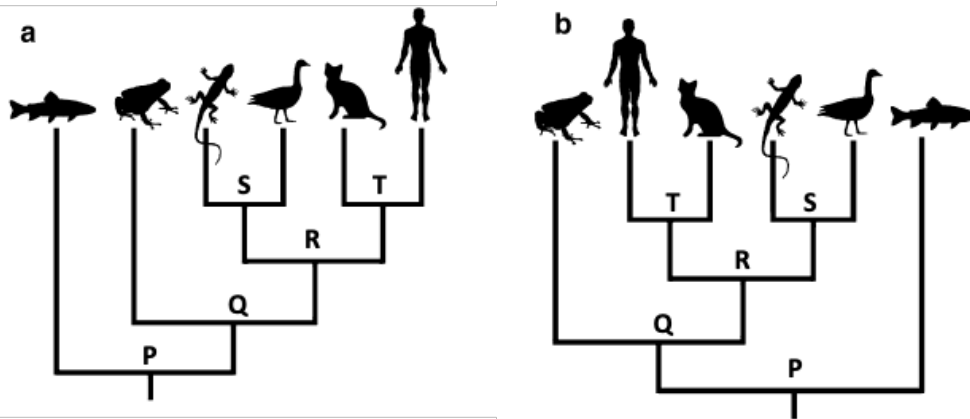


1. Consider the following Phylograms and answer the question with reasons.



- Which of the two trees (a) or (b) show the correct order of evolution from primitive to advanced forms? (2 marks)
 - Identify three clades in the tree (a) (3 marks)
 - Tree (b) shows that Frogs first evolved from sharks and then the lizards evolved from frogs. Comment on this statement. (2 marks)
2. Define (a) Red Queen Process (b) Adaptive Evolution (c) Disruptive Selection (3 marks)
3. Mexican tetras (*Astyanax mexicanus*) are fascinating fishes. They are found in the



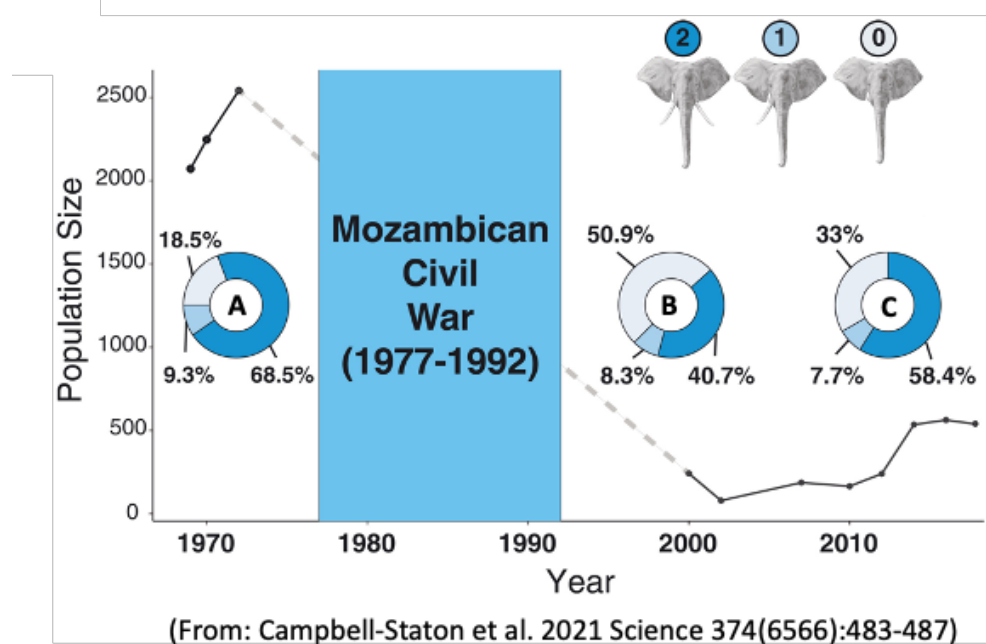
Richard Borowsky, Wikimedia Commons (CC BY-SA 3.0)

surface waters of lakes as well as in lakes inside deep dark caves. The two kinds of fish look dramatically different even though they are the same species. The surface fish have two, large eyes. The deep cave fish have no eyes. In addition, relative to surface fish, the cave fish are pale in colour, have smaller brain area related to

vision but larger brain area related to other sense, have a greater sense of “smell” and larger concentration of cells that detect vibrations in water. Using this example, comment on

- Fitness is context specific (2 marks)
 - Evolution is not “progressive” (2 marks)
 - Trade-offs are common (2 marks)
4. The ability to taste PhenylThioCarbamide (PTC) in humans is due to a single autosomal locus with two alleles. The ability to taste (T) is dominant. Suppose that from a population that is at Hardy-Weinberg Equilibrium for this trait, a random sample of 1000 individuals shows that 360 cannot taste PTC.
- Calculate the allelic and genotypic frequencies. (2+2 marks)
 - Which is the most common Phenotype? (1 mark)
 - Does the most common allele correspond to the most common phenotype? How do you explain this observation? (1 mark)

5. The Gorongosa National Park in Mozambique had a large population of African Elephants. In these elephants, all males have tusks. Among the females, some have two tusks, some have one tusk and others have no tusks. During the Mozambique civil war, the sale of ivory was used to finance the war and as a consequence, a large number of the tusked animals were killed for their tusks. The figure summarises the data. A, B, and C show the proportion of the three kinds of females in the population- before the war (A), the females that survived the war (B) and among the first generation of progeny born after the war.



- State the “four postulates” of evolution with reference to the figure. (2 marks)
 - Is this an example of Adaptive Evolution by Natural Selection? Back up your answer with reference to the figure. (2 marks)
 - What is the agent of selection in this case? (1 mark)
6. Consider three linked loci A-B-C (in that order). Let the genetic distance between A-B be 10cM and B-C be 20cM. If we sample 1000 gametes produced by a trihybrid individual (with respect to these three loci) how many of these gametes will be recombinants between A-B loci ONLY? (3 marks)